(1) By definition: $w_{t+1} = w_t - \eta_t g_t + \beta_t (w_t - w_{t-1})$, $w_0 = 0$, $\eta_0 = 0$.
$\forall\ t = 1, \dots, T$. The first formulation:

$$m_t = \hat{\beta}_t m_{t-1} + g_t\ ,\ w_{t+1} = w_t - \hat{\eta}_t m_t, m_0 = 0.$$

By this formulation:

$$w_{t-1} = w_t + \hat{\eta}_{t-1} m_{t-1} = w_t + \hat{\eta}_{t-1} \cdot (m_t - g_t) / \hat{\beta}_t.$$

$$\Rightarrow m_t = \frac{\hat{\beta}_t}{\hat{\eta}_{t-1}} (w_{t-1} - w_t) + g_t.$$

Finally: $w_{t+1} = w_t - \hat{\eta}_t \left(\frac{\hat{\beta}_t}{\hat{\eta}_{t-1}} (w_{t-1} - w_t) + g_t \right) =$

$= w_t - \hat{\eta}_t g_t + \frac{\hat{\eta}_t}{\hat{\eta}_{t-1}} \cdot \hat{\beta}_t (w_t - w_{t-1})$. So, for: $\hat{\eta}_t = \eta_t$,

$\hat{\beta}_t = \frac{\eta_{t-1}}{\eta_t} \beta_t$, this method reduces to $w_{t+1} = w_t - \eta_t g_t + \beta_t (w_t - w_{t-1})$.

Finally, for $w_0 = 0$, $\eta_0 = 0$, $w_1 = 0 = w_0 - \hat{\eta}_0 m_0 = 0$, so the two formulations are equivalent.

Next ↴

$$w_{t+1} = \frac{\bar{\beta}_{t+1}}{1+\bar{\beta}_{t+1}} w_t + \frac{1}{1+\bar{\beta}_{t+1}} v_t \; , \; v_t = v_{t-1} - \bar{\eta}_t g_t \; , \; v_0 = 0$$

By this formulation: $v_{t-1} = v_t + \bar{\eta}_t g_t$, $v_t = (1+\bar{\beta}_{t+1}) w_{t+1} - \bar{\beta}_{t+1} w_t$.

Then:

$$w_{t+1} = \frac{\bar{\beta}_{t+1}}{1+\bar{\beta}_{t+1}} w_t + \frac{1}{1+\bar{\beta}_{t+1}} \left(v_{t-1} - \bar{\eta}_t g_t \right) =$$

$$= \frac{\bar{\beta}_{t+1}}{1+\bar{\beta}_{t+1}} w_t + \frac{1}{1+\bar{\beta}_{t+1}} \left((1+\bar{\beta}_t) w_t - \bar{\beta}_t w_{t-1} - \bar{\eta}_t g_t \right) =$$

$$= \underbrace{\frac{\bar{\beta}_{t+1}}{1+\bar{\beta}_{t+1}} w_t + \frac{1}{1+\bar{\beta}_{t+1}} w_t}_{=1} - \frac{\bar{\beta}_t \bar{\eta}_t}{1+\bar{\beta}_{t+1}} g_t + \bar{\beta}_t \left(w_t - w_{t-1} \right) =$$

$$= w_t - \frac{\bar{\beta}_t \bar{\eta}_t}{1+\bar{\beta}_{t+1}} g_t + \bar{\beta}_t (w_t - w_{t-1}) .$$ So for : $\bar{\beta}_t = \beta_t$,

$\bar{\eta}_t = \frac{1+\beta_{t+1}}{\beta_t} \eta_t$, the formulation reduces to :

$$w_{t+1} = w_t - \frac{\bar{\beta} \bar{\eta}_t}{1+\bar{\beta}_{t+1}} g_t + \bar{\beta}_t (w_t - w_{t-1}) = w_t - \eta_t g_t + \beta_t (w_t - w_{t-1}) .$$

Finally for $w_0 = 0$, $\eta_0 = 0$, $v_0 = 0$, $w_1 = 0 = \frac{\bar{\beta}_1}{1+\bar{\beta}_1} \underset{0}{w_0} + \frac{1}{1+\bar{\beta}_1} v_0 = 0$, so the two formulations are equivalent.

So, this shows both formulations are equivalent.

(2) By definition in class:

$$e_{t+1} = m_{t+1} - \nabla f(w_{t+1}) \underbrace{=}_{\text{def of } m_{t+1}} \nabla l(w_{t+1}, z_{t+1}) + \beta \overbrace{(m_t}^{= e_t + \nabla f(w_t)} - \nabla l(w_t, z_{t+1}) - \nabla f(w_{t+1})$$

$$= \nabla l(w_{t+1}, z_{t+1}) + \beta(e_t + \nabla f(w_t) - \nabla l(w_t, z_{t+1})) - \nabla f(w_{t+1}) =$$

$$= \nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1}) + \beta e_t + \beta(\nabla f(w_t) - \nabla l(w_t, z_{t+1})) \underbrace{=}_{\text{add and subtract } \beta(\nabla f(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1})) \text{ by hint}}$$

$$= \overbrace{(1-\beta)(\nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1}))}^{b} + \overbrace{\beta e_t}^{a} + \overbrace{\beta(\nabla f(w_t) - \nabla f(w_{t+1}) + \nabla l(w_{t+1}, z_{t+1}) - \nabla l(w_t, z_{t+1}))}^{c}$$

Using this result (rearranging, $\|a+b+c\|^2$

$$\|e_{t+1}\|^2 = \beta^2 \|e_t\|^2 + (1-\beta)^2 \|\nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1})\|^2 +$$
$$+ \beta^2 \|\nabla f(w_t) - \nabla f(w_{t+1}) + \nabla l(w_{t+1}, z_{t+1}) - \nabla l(w_t, z_{t+1})\|^2$$
$$+ 2\beta(1-\beta)\langle e_t, \nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1})\rangle$$
$$+ 2\beta^2 \langle e_t, \nabla f(w_t) - \nabla f(w_{t+1}) + \nabla l(w_{t+1}, z_{t+1}) - \nabla l(w_t, z_{t+1})\rangle$$
$$+ 2\beta(1-\beta)\langle \nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1}), \nabla f(w_t) - \nabla f(w_{t+1}) + \nabla l(w_{t+1}, z_{t+1}) - \nabla l(w_t, z_{t+1})\rangle.$$

Finally, taking $\mathbb{E}[\cdot | w_t]$ (total expectation):

$$\mathbb{E}[\langle e_t, \nabla l(w_{t+1}, z_{t+1}) - \nabla f(w_{t+1})\rangle | w_t] = \langle e_t, \overbrace{\mathbb{E}[\nabla l(w_{t+1}, z_{t+1}) | w_t]}^{\nabla f(w_{t+1})} - \nabla f(w_{t+1})\rangle =$$
$$= \langle e_t, \nabla f(w_{t+1}) - \nabla f(w_{t+1})\rangle = \langle e_t, 0\rangle = 0$$

and similarly:

$$\mathbb{E}[\langle e_t, \nabla f(w_t) - \nabla f(w_{t+1}) + \overbrace{\nabla l(w_{t+1}, z_{t+1})}^{\mathbb{E}[\nabla l(w_{t+1}, z_{t+1}) | w_t] = \nabla f(w_{t+1})} - \underbrace{\nabla l(w_t, z_{t+1})}_{\nabla f(w_t).}\rangle | w_t] =$$
$$= \langle e_t, \nabla f(w_t) - \nabla f(w_{t+1}) + \nabla f(w_{t+1}) - \nabla f(w_t)\rangle = \langle e_t, 0\rangle = 0$$

So: ↴

$$\mathbb{E}\left[\|e_{t+1}\|^2 \mid w_t\right] \le \beta^2\|e_t\|^2 + 2(1-\beta)^2 \underbrace{\mathbb{E}\left[\|\nabla L(w_{t+1},z_{t+1}) - \nabla f(w_{t+1})\|^2 \mid w_t\right]}_{\le \sigma^2} +$$

$$+2\beta^2 \mathbb{E}\left[\|\nabla f(w_t) - \nabla f(w_{t+1}) + \nabla L(w_{t+1},z_{t+1}) - \nabla L(w_t,z_{t+1})\|^2 \mid w_t\right] \le$$

(where I distributed the $\beta(1-\beta)$ in the last cross term and used $\mathbb{E}[\langle b,c\rangle \mid w_t] \le \mathbb{E}[\|b\|^2 \mid w_t] + \mathbb{E}[\|c\|^2 \mid w_t]$ and then grouped the terms).

$$\le \beta^2\|e_t\|^2 + 2(1-\beta)^2\sigma^2 + 2\beta^2 \underbrace{\mathbb{E}\left[\|(\nabla L(w_{t+1},z_{t+1}) - \nabla L(w_t,z_{t+1})) - (f(w_{t+1}) - \nabla f(w_t))\|^2 \mid w_t\right]}$$

$\le$ by SFO variance assumption

$\mathbb{E}[\|X - \mathbb{E}[X]\|^2] \rightleftharpoons \mathbb{E}[\|X\|^2] - 2\langle \mathbb{E}[X], \mathbb{E}[X]\rangle + (\mathbb{E}[X])^2 = \mathbb{E}[\|X\|^2] - (\mathbb{E}[X])^2 \le \mathbb{E}[X^2]$, holds for $\mathbb{E}[\|X - \mathbb{E}[X \mid w_t]\|^2 \mid w_t]$ too.

$$\le \beta^2\|e_t\|^2 + 2(1-\beta)^2\sigma^2 + 2\beta^2\mathbb{E}\left[\|\nabla L(w_{t+1},z_{t+1}) - \nabla L(w_t,z_{t+1})\|^2 \mid w_t\right]$$

$$\underset{w}{\le} \beta^2\|e_t\|^2 + 2(1-\beta)^2\sigma^2 + 2\beta^2 L^2 \mathbb{E}\left[\|w_{t+1} - w_t\|^2 \mid w_t\right] \underset{w}{=}$$

L is L-lipschitz η

$w_{t+1} - w_t = -\eta m_t$

$$= \beta^2\|e_t\|^2 + 2(1-\beta)^2\sigma^2 + 2\beta^2 L^2\eta^2 \mathbb{E}\left[\|m_t\|^2 \mid w_t\right].$$ Take total exp

Taking total exp, and let $\lambda = 1-\beta$, then replace λ with new β (basically $m_t = \nabla L(w_t,z_t) + (1-\beta)(m_{t-1} - \nabla L(w_{t-1},z_t))$):

$$\mathbb{E}\|e_{t+1}\|^2 \le \beta^2\|e_t\|^2 + 2(1-\beta)^2\sigma^2 + 2\beta^2 L^2\eta^2 \mathbb{E}\|m_t\|^2$$ or for the $1-\beta$ formulation of m_t:

$$\mathbb{E}\|e_{t+1}\|^2 \le (1-\beta)^2\|e_t\|^2 + 2\beta^2\sigma^2 + 2(1-\beta^2)L^2\eta^2\mathbb{E}\|m_t\|^2,$$

proving the inequality. □

(3) By definition $e_{t+1} = (g_t + e_t) - C(g_t + e_t)$, so:

$$\mathbb{E}_{C,z_t}\left[\|e_{t+1}\|^2 \mid w_t\right] \underset{\text{LOTE and def of } e_{t+1}}{=} \mathbb{E}_{z_t}\left[\mathbb{E}_C\left[\|\overbrace{C(g_t+e_t)}^{\text{fixed}} - (g_t+e_t)\|^2 \mid g_t, e_t\right] \mid w_t\right]$$

$$\leq \mathbb{E}_{z_t}\left[(1-\delta)\|g_t+e_t\|^2 \mid w_t\right] = (1-\delta)\,\mathbb{E}_{z_t}\left[\|g_t+e_t\|^2 \mid w_t\right].$$

by the assumption that the compression mechanism is contractive, so $\mathbb{E}\|C(x)-x\|^2 \leq (1-\delta)\,\mathbb{E}\|x\|^2$ where $x = g_t+e_t$, $C(x) = C(g_t+e_t)$, also $\mathbb{E}[\|x\|^2 \mid x] = \|x\|^2$

So:

$$\mathbb{E}_{C,z_t}\left[\|e_{t+1}\|^2 \mid w_t\right] \leq (1-\delta)\,\mathbb{E}_{z_t}\left[\|g_t+e_t\|^2 \mid w_t\right] \underset{\text{Youngs}}{\leq}$$

$$\leq (1-\delta)\,\mathbb{E}_{z_t}\left[(1+\lambda)\|e_t\|^2 + (1+\lambda^{-1})\|g_t\|^2 \mid w_t\right] \underset{\text{choose } \lambda=\delta}{=}$$

$$= \mathbb{E}_{z_t}\left[\underbrace{(1-\delta^2)}_{=(1+\delta)(1-\delta)}\|e_t\|^2 + \underbrace{\left(\frac{1-\delta^2}{\delta}\right)}_{\leq \frac{1}{\delta}}\|g_t\|^2 \mid w_t\right] \leq$$

$$\leq (1-\delta^2)\|e_t\|^2 + \frac{1}{\delta}\,\mathbb{E}\left[\|g_t\|^2 \mid w_t\right] \leq$$

$e_t \perp z_t$; $\frac{1-\delta^2}{\delta} \leq \frac{1}{\delta}$; $\mathbb{E}\left[\|g_t\|^2 \mid w_t\right] \leq \sigma^2 + \|\nabla f(w_t)\|^2$ by SFO assumption; by SFO assum and $\mathbb{E}\left[\|\nabla f(w_t)\|^2 \mid w_t\right] = \|\nabla f(w_t)\|^2$

$$\leq (1-\delta^2)\|e_t\|^2 + \frac{1}{\delta}(\sigma^2 + \|\nabla f(w_t)\|^2).$$ Multiplying out and distributing $\frac{1}{\delta}$, I get:

$$\mathbb{E}_{C,z_t}\left[\|e_{t+1}\|^2 \mid w_t\right] \leq (1-\delta^2)\|e_t\|^2 + \frac{\|\nabla f(w_t)\|^2}{\delta} + \frac{\sigma^2}{\delta},$$

proving the desired bound. □